PaperRank

DCV Course Project Report

Live dashboard: https://paperrank-d7ql.onrender.com/

Code repository: https://github.com/kushal-in/paperrank

Member	Roll number	Contribution
Bhanu Manideep	DA25B024	ROR and OpenAlex collection, data cleaning
Sasanka Mouli	DA25B013	NIRF scraping, data cleaning
Kushal V	DA25B019	Visualisation and dashboard

Question: Does the NIRF research score track independent publication and citation counts from OpenAlex?

Answer in brief: Mostly yes. Across 164 institutions and nine ranking years (2017 to 2025), the NIRF research score agrees closely with OpenAlex citations (Spearman 0.86 in 2025, between 0.84 and 0.92 in every year). It agrees less with paper counts (0.72 in 2025, down from 0.89 in 2017), and for institutions ranked 51 to 100 the agreement with paper counts fell from 0.88 to 0.27.

1. Introduction

Every year NIRF (the National Institutional Ranking Framework) ranks India's top 100 institutions, and part of the score rewards research. We asked whether an independent count of papers and citations orders institutions the same way the NIRF research score does.

Objectives

- Collect NIRF Overall rankings for 2017 to 2025 and independent publication data from OpenAlex.
- Clean both sources, link them through ROR ids, and store a documented clean table.
- Measure agreement with Spearman rank correlation per year, with 95% bootstrap intervals.
- Present the result in an interactive dashboard.

Scope. 164 institutions, 9 ranking years, 901 institution-years. Each NIRF year Y is compared with OpenAlex output over the three years before it (Y−3 to Y−1), using the institution's own papers.

What this report covers. Collection (section 2), cleaning (3), the clean table and data dictionary (4), analysis (5), the dashboard (6), findings and limitations (7), and how to rerun the project (8).

2. Data collection

We used three sources. Notebooks 1, 3 and 4 collect them; every download is saved untouched in `raw/`.

Source	What we took	Licence or terms	Collected
NIRF Overall ranking pages, nirfindia.org/ Rankings/YEAR/ OverallRanking.html	Rank and five component scores (TLR, RPC, GO, OI, PR) for 2017 to 2025	Public government pages; no robots.txt and no terms page (see below)	2026-10-31
OpenAlex API, api.openalex.org	Papers and citations per institution and year; institution records	CC0 (public domain)	2026-11-01
ROR API v2, api.ror.org	Registry ids that link NIRF names to OpenAlex records	CC0 1.0	2026-11-01

Method

1. **Notebook 1, NIRF.** Fetched the nine yearly pages (one request at a time, 1.5 seconds apart) and parsed rank, name, city and scores into one table.
2. **Notebook 3, ROR matching.** Searched ROR for each of the 164 institutions and scored the candidates on name and city similarity. 157 matches were accepted automatically and 7 were decided by hand (`ror_manual.csv`).
3. **Notebook 4, OpenAlex.** For each ROR id, fetched the institution record and yearly counts of works: all works, articles, works with at least 1 citation, works with at least 10 citations, own works only, and works including units filed beneath the institution (lineage). Answers are cached as JSON in `raw/openalex/`.

Ethics and terms. NIRF's server returned a 404 page for robots.txt on 2026-10-31, so no crawl rules are published. The site has no terms page. We treat the tables as public records, use them for non-commercial academic work, name NIRF as the source and do not redistribute them. OpenAlex and ROR were called through their public APIs with pauses between calls.

Raw files are never edited. Every later step reads from `raw/` and writes to `interim/` or `clean/`. Raw data totals about 29 MB, under the 50 MB limit.

3. Data cleaning

Cleaning merged duplicate spellings and left every row in place: 901 NIRF rows before and 901 after. The full log is `interim/cleaning_log_final_2026-11-02.csv` (22 entries), written by notebook 7 from notebook 2's `cleaning_log_2026-11-02.csv`.

Measure	Before	After
Distinct NIRF institution ids	342	164
Distinct name spellings	231	164
Distinct city spellings	106	92
NIRF rows	901	901

Identity. NIRF changed its id format in 2017, so the same institution had two ids. 85 institution-years from 2017 were matched to a later id by name (9 via an alias table), 15 institutions keep a placeholder id (`X-####`), and 2 display names were fixed by hand. 43 city labels were standardised; the raw label is kept in `location_raw`.

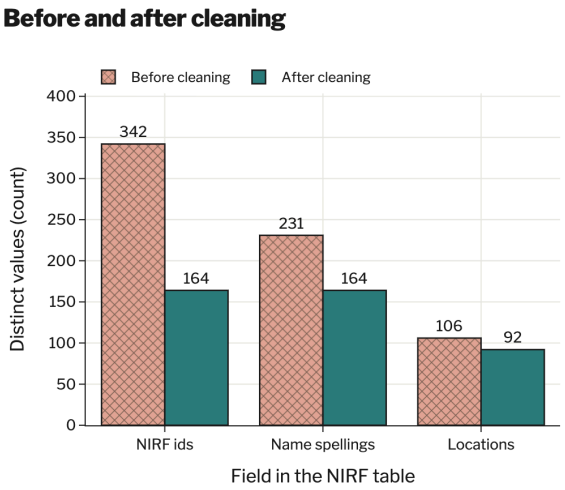
Missing values. No NIRF score or OpenAlex window count is empty. 5,321 empty OpenAlex yearly counts are structural (5,311 are years before 2010, when OpenAlex's yearly counts begin; 10 are future years) and lie outside every window. 90 institution-years vanish from the next year's top 100, and they ranked lower (mean rank 81.1 against 46.7, permutation p = 0.0001). The gap depends on observed rank, so we treat it as MAR, keep observed rows, and impute nothing. 7 perception scores are exactly 0.0 (all at ranks 67 to 97); we kept them as published because a true 0 cannot be told from an unreported value, and perception is not used in the main result.

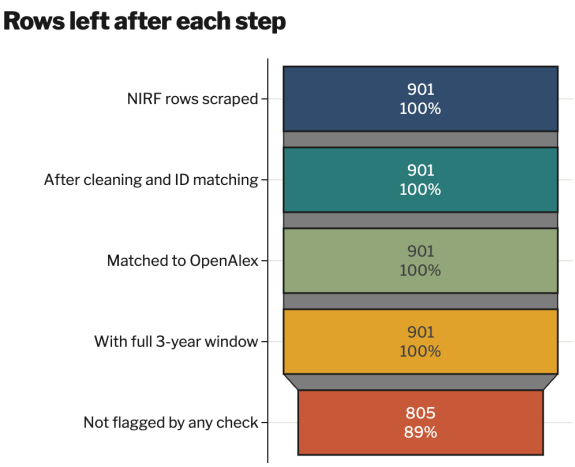
Outliers. Three methods (IQR on log papers, modified z-score, Mahalanobis distance) flagged 7 institution-years: 4 with very few papers (specialist or management institutes) and 3 with very many (large research universities). All are real, so we kept them and reran the main result without them (`excl_flagged` in the Spearman table).

Duplicates. 0 exact repeats and 0 cases of two institutions sharing one OpenAlex id. 8 pairs of near-identical names (for example IIT campuses and AIIMS sites) were checked by eye and are different places. 15 NIRF-versus-OpenAlex name mismatches were checked against ROR city and type and are renamed or abbreviated institutions, not wrong matches.

Paper-count correction. OpenAlex's broader (lineage) count includes affiliated units. It exceeded the institution's own count by more than 5% in 88 institution-years, so we use the own count as the main measure. For the 2025 window this lowers the total from 662,956 to 642,584 papers (3.1% fewer). Flags: `flag_lineage_inflated` 88, `flag_oa_disagree` 4, `flag_aiims_merged` 4.

Join checks (join_log). All pass: 164 crosswalk rows, 0 shared OpenAlex ids, 0 NIRF rows without an OpenAlex id, 0 rows with missing or fewer than three window years.





4. Clean table and data dictionary

The clean table is `clean/nirf_openalex_joined_2026-11-02.csv`: 901 rows (one per institution and ranking year) and 40 columns. A copy is stored in `data/dcv_2026-11-02.sqlite`; reading it back gives the same shape, values, empty cells and dtypes as the CSV. The full dictionary, one row per column with dtype, unit, source, allowed range and observed range, is `clean/data_dictionary.csv`. The groups below summarise it.

Group	Columns	Meaning
Identity	`year`, `id_new`, `name_new`, `location_raw`, `location_clean`	Ranking year (2017 to 2025), our institution id, one lower-case name per institution, city as printed and city after spelling fixes
NIRF scores	`rank`, `tlr`, `rpc`, `go`, `oi`, `pr`	Overall rank (1 = best, ties share a number) and the five component scores, 0 to 100. `rpc` is the research score used in the analysis
Linking	`openalex_id`, `ror_id`, `decision`	OpenAlex institution id, ROR id, and whether the match was `auto` or `manual`
OpenAlex counts (3-year window)	`pubs_own`, `pubs_lineage`, `pubs_article_lineage`, `cited1_lineage`, `cited10_lineage`, `oa_pubs_year`, `cites_received`	Papers and citations in the three years before the ranking year, under different definitions
Window	`n_oa_years`, `n_oa_missing`, `win_from`, `win_to`	Years covered by the window and completeness checks (3 years, 0 missing for every row)
Sensitivity	`pubs_own_shift1`, `cites_received_shift1`	The same counts with the window moved back one year
Derived measures	`pubs_per_year`, `share_ge10_lineage`, `lineage_ratio`, `oa_ratio`	Papers per year, share of papers with at least 10 citations, and ratios between counting methods
Flags	`flag_lineage_inflated`, `flag_oa_disagree`, `flag_aiims_merged`, `flag_any`	Rows where counting methods disagree by more than a set threshold
Rank comparison	`rank_rpc`, `rank_pubs_own`, `rank_cites_received`, `rank_gap_pubs_vs_rpc`, `rank_gap_cites_vs_rpc`	Within-year ranks and the gap between the research-score rank and the output rank

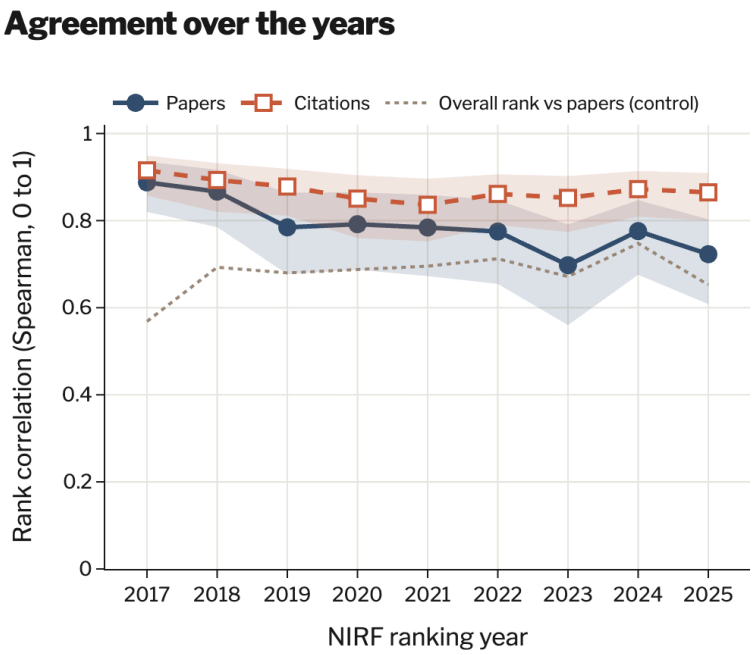
All 40 columns are described in the dictionary, and every observed range lies inside its allowed range (checked in notebook 7).

5. Analysis and visualisation

Notebook 6 joins the two sources and computes, for each ranking year, the Spearman rank correlation between the NIRF research score (rpc) and each OpenAlex measure, with a 95% bootstrap interval. A value of 1 means the same order and 0 means unrelated. Results are in `interim/`

spearman_by_year_2026-11-02.csv (seven specifications: main, excluding flagged rows, top 50 only, ranks 51 to 100, shifted window, lineage count, and a control for overall rank).

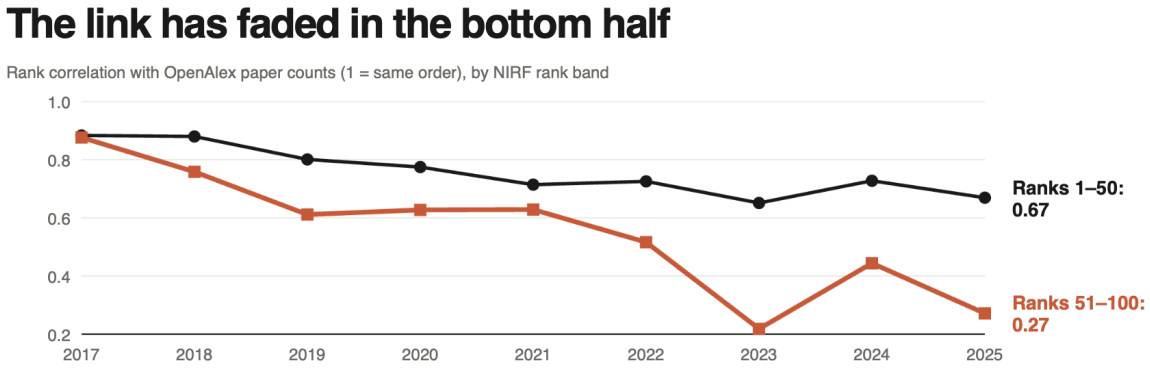
Citations track the research score closely. The correlation with OpenAlex citations is 0.86 in 2025



(95% interval 0.80 to 0.91) and stays between 0.84 and 0.92 in all nine years.

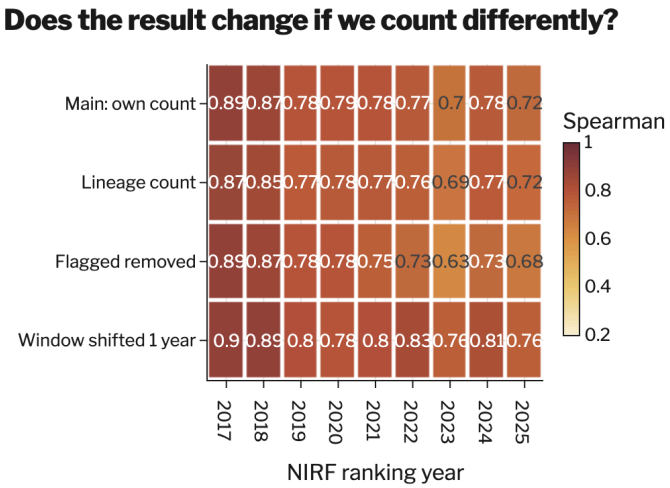
Paper counts track it less, and the link is loosening. The correlation with paper counts was 0.89 in 2017 and is 0.72 in 2025 (interval 0.61 to 0.80), with a dip to 0.70 in 2023.

The weakening is in the bottom half of the list. For ranks 51 to 100, agreement with paper counts fell from 0.88 in 2017 to 0.27 in 2025 (low of 0.22 in 2023); for ranks 1 to 50 it stayed at 0.65 or higher. The share of highly cited papers (10 or more citations) matches the research score at only 0.44 in 2025.



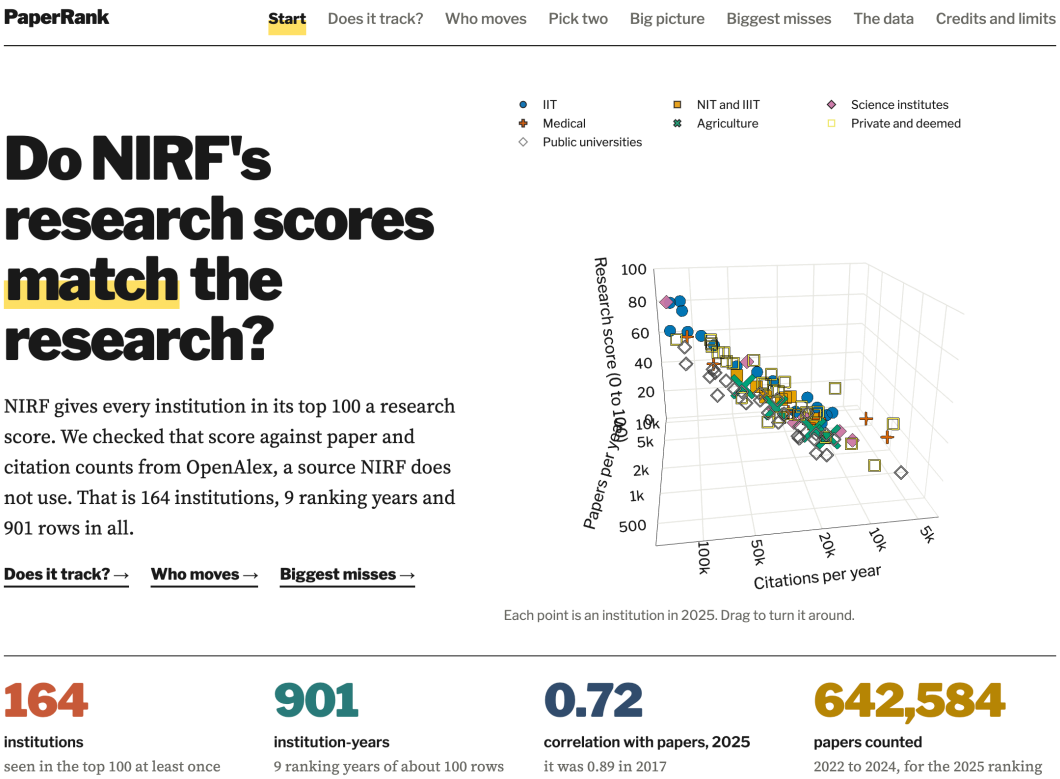
Biggest single miss. In 2017 the research score put JNCASR 11th, yet it was 60th of 100 on papers.

Robustness. Moving the three-year window back one year changes any correlation by at most 0.06. Dropping the flagged rows moves any correlation by at most 0.07 and leaves the pattern intact (for example, paper counts in 2025 go from 0.72 to 0.68). See the `excl_flagged` and `shifted_window` rows of the Spearman table.



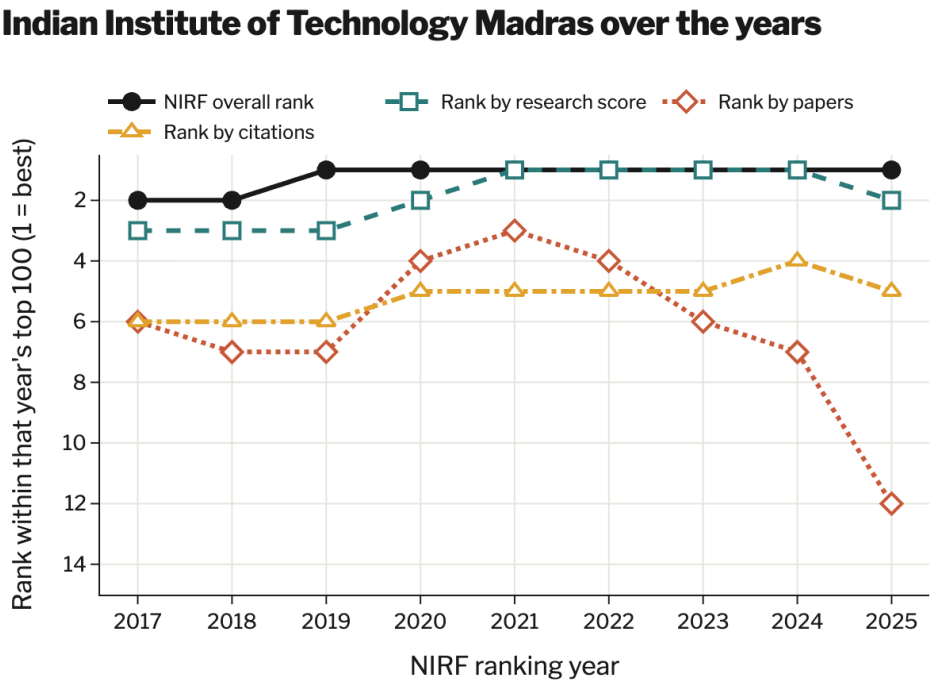
6. Dashboard

The dashboard (folder `retro5/`) is a Plotly Dash app with an essay-style layout. It reads the clean table, the Spearman results and the cleaning logs from `clean/` and `interim/`, and queries the SQLite file in `data/` with parameterised SQL. Live link: https://paperrank-d7ql.onrender.com/



Tab	What it shows
Front page	Spinning 3D scatter of research score against output, ranking race, Sankey
Does it track?	Animated scatter, gauges, correlation by year, robustness checks
Flows	Two Sankeys, treemap, sunburst
Institutions	Radar, trajectories, bump chart, rank calendar
Landscape	Violin, 3D surface, parallel coordinates
Gaps	Lollipop of rank gaps, churn, violin
Behind the data	Before/after correction, funnel, dumbbell, outliers, cleaning log
Credits	Team, sources, licences and terms, libraries, AI tools

Design choices. One highlight colour against grey; a "what it hides" note under each chart; a shared selection that links the charts; and a colour-blind simulation check (`python cbcheck.py`).



Its NIRF rank and its rank by papers differ by up to 11 places in a single year. The solid line is NIRF's rank; the others come from our counts.

What it hides. Ranks exist only inside the top 100, so a missing point means outside the list, not a bad rank.

Run locally.

```
pip install -r requirements.txt
cd retro5
python app.py   # then open http://127.0.0.1:8050
```

7. Findings, limitations and conclusion

Does the NIRF research score track independent output? Mostly yes, and best at the top.

Measure	Result
Agreement with OpenAlex citations, 2025	0.86 (95% interval 0.80 to 0.91); 0.84 to 0.92 in every year
Agreement with paper counts, 2025	0.72 (0.61 to 0.80), down from 0.89 in 2017
Agreement with paper counts, ranks 51 to 100, 2025	0.27, down from 0.88 in 2017
Agreement with paper counts, ranks 1 to 50, 2025	0.67 (0.65 or higher in every year)

The research score is a dependable guide to who sits near the top. It says much less about the bottom half of the list, and the link has loosened since 2017.

Limitations

1. **Agreement is not cause.** Rank correlation shows the orders match, not that publications drive the score. The NIRF research score also counts patents and funded projects, so it was never meant to equal publications.
2. **Only institutions on the list.** Institutions that drop out ranked lower (mean rank 81.1 against 46.7), so the results hold for the top 100, not for all Indian institutions.
3. **Matching names to records.** OpenAlex counts can include affiliated units. We used the own-papers count and flagged 88 institution-years where the broader count was more than 5% higher.
4. **Window choice.** We used three years before each ranking year. Shifting it by one year moves any correlation by at most 0.06, so the story holds but exact values depend on it.
5. **A snapshot.** Data was collected between 2026-10-31 and 2026-11-02. OpenAlex keeps adding records, and 7 perception scores of exactly 0 are kept as published.

Next steps. Compare field-normalised citations, and test each NIRF score component against its own OpenAlex counterpart.

Credits, licences and AI use. NIRF pages (public, used for academic work), OpenAlex (CC0), ROR (CC0 1.0). Libraries: pandas, NumPy, SciPy, Requests, Beautiful Soup, SQLite, Plotly, Dash, dash-bootstrap-components, Jupyter. Claude (Anthropic) helped draft and review scripts for scraping, cleaning, matching, analysis and the dashboard; the team ran every script, checked the outputs and made the decisions.

8. Submission contents and how to reproduce

```
paperrank_submission/
  README.md                how the project fits together, plus a map of required items
  requirements.txt
  .env.example             OpenAlex key goes here (copy to .env)
  ror_manual.csv           7 hand decisions for ROR matches
  notebooks/               01_collect_nirf ... 07_clean_check_and_store
```

```
raw/          NIRF pages, OpenAlex JSON cache (untouched)
interim/      parsed tables, crosswalks, tests, cleaning logs
clean/        nirf_openalex_joined, nirf_overall_clean, institutions, data_dictionary.csv
data/         dcv_2026-11-02.sqlite (the joined table)
retro5/       the dashboard (python app.py)
report/       this report as PDF and presentation slides
```

Order of the notebooks

#	Notebook	Needs internet	Output
1	01_collect_nirf	yes	`raw/` NIRF pages, `interim/nirf_overall_scraped`
2	02_build_clean_nirf	no	`clean/nirf_overall_clean`, `clean/institutions`, first cleaning log
3	03_match_ror	yes	`interim/ror_candidates`, `interim/ror_best`
4	04_collect_openalex	yes, key in `.env`	`raw/openalex/`, `interim/openalex_*`
5	05_profile_raw_files	no	shape, dtypes, empty cells and meaning of every column
6	06_join_analyse	no	`clean/nirf_openalex_joined`, `interim/spearman_by_year`, `join_log`
7	07_clean_check_and_store	no	`interim/cleaning_log_final`, `clean/data_dictionary.csv`, `data/*.sqlite`

Each step reads the newest dated file in the previous folder. Notebooks 2, 5, 6 and 7 work offline from the files supplied and reproduce the same clean table, Spearman results and logs. Notebooks 1, 3 and 4 call websites and are included with their saved outputs; a trial run of notebook 4 (`--limit 3`) writes a small newer crosswalk that breaks step 6, so delete the files a trial writes.